PRESBURGER LOGIC AND SEMI-LINEAR SETS

Ayush Priyadarshi, Anirudh Gupta, Aditya Arsh, Yash Kamble

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Indian Institute of Science, Banglore

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What is Presburger Arithmetic?

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Presburger arithmetic is the first-order theory of the natural numbers with addition, named after Mojżesz Presburger, who introduced it in 1929.

- The language includes addition and equality but omits multiplication.
- It is strictly weaker than Peano arithmetic, which includes both addition and multiplication.

2.2

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- The language includes addition and equality but omits multiplication.
- It is strictly weaker than Peano arithmetic, which includes both addition and multiplication.
- All formulas in the theory are decidable.
- Presburger Arithmetic is decidable while Peano Arithmetic is not.

Axioms of Presburger Arithmetic

Universal Axioms

The axioms of Presburger arithmetic are the universal closures of the following formulas:

1. $\neg(0 = x + 1)$
2. $x + 1 = y + 1 \rightarrow x = y$
3. $x + 0 = x$
4. $x + (y + 1) = (x + y) + 1$
5. Induction Schema: Let $P(x)$ be any formula in the language of Presburger arithmetic (with x free). Then:

$$(P(0) \wedge \forall x\,(P(x) \rightarrow P(x + 1))) \rightarrow \forall y\, P(y)$$

4.1 First-Order Logic (FOL)

What is First-Order Logic?

First-order logic (FOL), also called predicate logic, extends propositional logic by allowing:

- Quantified variables: $\forall x$, $\exists y$, etc.
- Predicates for properties and relations: e.g., $\text{Man}(x)$, $\text{Mortal}(x)$

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- Predicates for properties and relations: e.g., $\mathrm{Man}(x)$, $\mathrm{Mortal}(x)$
- Logical sentences like:

$$\forall x(\mathrm{Man}(x) \rightarrow \mathrm{Mortal}(x))$$

FOL vs Second-Order Logic

Quantification in First-Order Logic

In FOL, we can quantify over individuals, but not over predicates:

$$\exists x\, \mathrm{Cube}(x) \quad \checkmark\text{Valid in FOL}$$

$$\exists P\, P(b) \quad \text{Not valid in FOL (only in SOL)}$$

In Second-Order Logic (SOL), predicates can be quantified:

$$\exists P \forall x (Px \leftrightarrow (\mathrm{Cube}(x) \vee \mathrm{Tet}(x)))$$

Presburger Logic

First-Order Logic over $(\mathbb{N}, 0, 1, +, =, <)$

- Interpreted over $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$
- Allowed operations: addition $(+)$, order $(<)$, and logical connectives:

$$\neg, \wedge, \vee, \forall, \exists$$

- You can express:
  - $\forall x \forall y((x < y) \Rightarrow \exists z(x < z < y))$
  - "Every number is odd or even":

$$\forall x \exists y(x = 2y \vee x = 2y + 1)$$

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Presburger Logic Limitations

What You **Cannot** Express

- Multiplication of variables (non-linear terms), e.g.:

$$x \cdot y, \quad x^2$$

- Properties like:
  - Primality: prime(x)
  - Divisibility: $y \mid x$

7.2

Presburger Logic Limitations

What You **Cannot** Express

- Multiplication of variables (non-linear terms), e.g.:

$$x \cdot y, \quad x^2$$

- Properties like:
  - Primality: prime(x)
  - Divisibility: $y \mid x$
- These require first-order arithmetic with multiplication.

Presburger Logic: Syntax and Semantics

Terms and Formulas

Terms t are built using:

$$t ::= 0 \mid 1 \mid x \mid y \mid t + t$$

Atomic formulas f:

$$f ::= t = t \mid t < t$$

General formulas φ:

$$\varphi ::= f \mid \neg\varphi \mid \varphi \vee \varphi \mid \varphi \wedge \varphi \mid \exists x\, \varphi \mid \forall x\, \varphi$$

Semantics: $L(\varphi)$ is the set of all interpretations (assignments to variables) that make φ true.

9.1 What is Decidability?

Formal Definition

A set $S \subseteq \Sigma^*$ is decidable if there exists a Turing machine (or equivalently, an algorithm) that halts on every input and:

- Accepts inputs $x \in S$
- Rejects inputs $x \notin S$

9.2

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Intuitive Idea

There is a finite procedure that always terminates and determines whether a given input belongs to the set or not.

Peano Arithmetic is Undecidable

A Natural Decision Problem

Given a polynomial equation with integer coefficients:

$$p(x_1, x_2, \ldots, x_n) = 0$$

Does it have a solution in $\mathbb{N}$?

10.2 Peano Arithmetic is Undecidable

A Natural Decision Problem

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Does it have a solution in $\mathbb{N}$?

- $x^2 + y^2 - 1 = 0$ Yes

10.3 Peano Arithmetic is Undecidable

A Natural Decision Problem

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Does it have a solution in $\mathbb{N}$?

- $x^2 + y^2 - 1 = 0$ Yes
- $x^2 + y^2 - 3 = 0$ No

Peano Arithmetic is Undecidable

A Natural Decision Problem

Given a polynomial equation with integer coefficients:

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Does it have a solution in $\mathbb{N}$?

- $x^2 + y^2 - 1 = 0$ Yes
- $x^2 + y^2 - 3 = 0$ No
- $(x+2)^2 - 61y^2 = 1$

Smallest Solution Example

Smallest Solution

Consider the equation:

$$(x+2)^2 - 61y^2 = 1$$

The smallest solution in $\mathbb{N}$ is:

$$x = 1766319047, \quad y = 226123980$$

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Hilbert's Tenth Problem

The Problem

a polynomial equation with integer coefficients:

$$p(x_1, x_2, \ldots, x_n) = 0$$

Is there an algorithm that can decide whether it has a solution in $\mathbb{N}$ or $\mathbb{Z}$?

Hilbert's Tenth Problem

The Problem

a polynomial equation with integer coefficients:

$$p(x_1, x_2, \ldots, x_n) = 0$$

Is there an algorithm that can decide whether it has a solution in $\mathbb{N}$ or $\mathbb{Z}$?

Answer (Matiyasevich, 1970)

No. There is no such algorithm.
The problem was solved in the negative.

Therefore, Peano Arithmetic is undecidable.

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Decision Problem for Presburger Arithmetic

Key Question

Does there exist a finite procedure (algorithm) that, given a statement in Presburger arithmetic, always decides whether it is true or false?

13.2 Decision Problem for Presburger Arithmetic

Key Question

Does there exist a finite procedure (algorithm) that, given a statement in Presburger arithmetic, always decides whether it is true or false?

Answer

- **Yes!** Presburger arithmetic is decidable.

Automata-Based Construction

Encoding Variable

Interpret each assignment as a binary string:

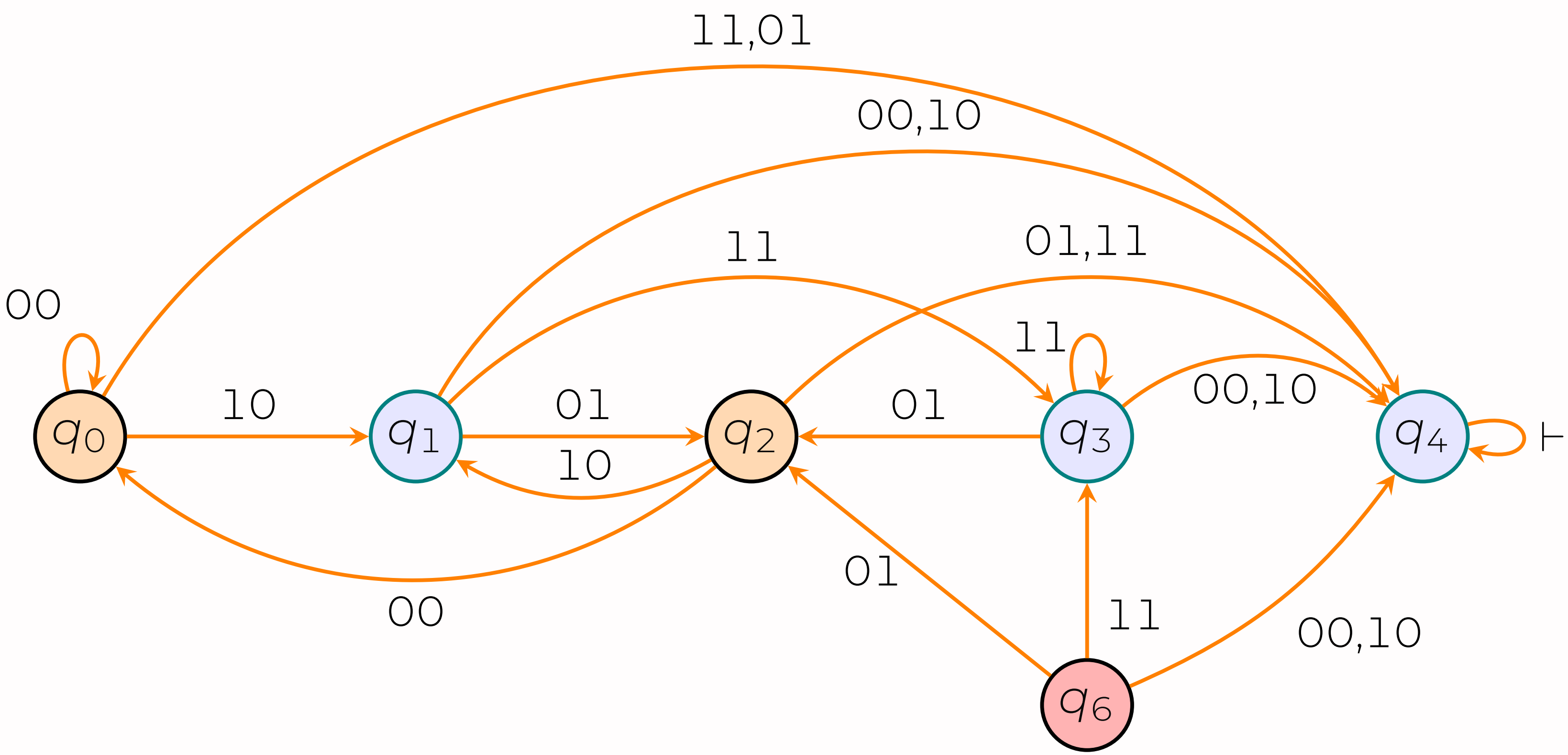
$$x \mapsto 00101 \qquad y \mapsto 00010$$

Each assignment corresponds to a row in a binary word.

Automata for Atomic Formulas

Construct finite automata over such binary words that accept all satisfying assignments for atomic formulas.

Construction of automata for $y = 2x + 1$



accepting state $\rightarrow$ *orange*
initial state $\rightarrow$ *red*

Construction for Atomic Formulas

Consider the formula φ: $a_1 x_1 + \cdots + a_n x_n = b$, where $a_i \in \mathbb{Z}$:

Construct automaton $\mathcal{A}_\varphi$ as follows:

- Begin with the initial state labelled b.
- In general, if the current state is c, and the input is a bit vector $(\theta_1, \ldots, \theta_n)$:
  - Check if $(a_1\theta_1 + \cdots + a_n\theta_n) \equiv c \pmod 2$.
  - If so, move to the state labelled

$$\frac{c - (a_1\theta_1 + \cdots + a_n\theta_n)}{2}.$$

  - Otherwise, move to the Junk state.
- Make the state labelled 0 the (only) final state.

Claim on State Bound

The number of states is bounded by $2M + 2$ where

$$M = \max(|b|, |a_1| + \cdots + |a_n|).$$

Example 2

$2x + y - 3z = 2$

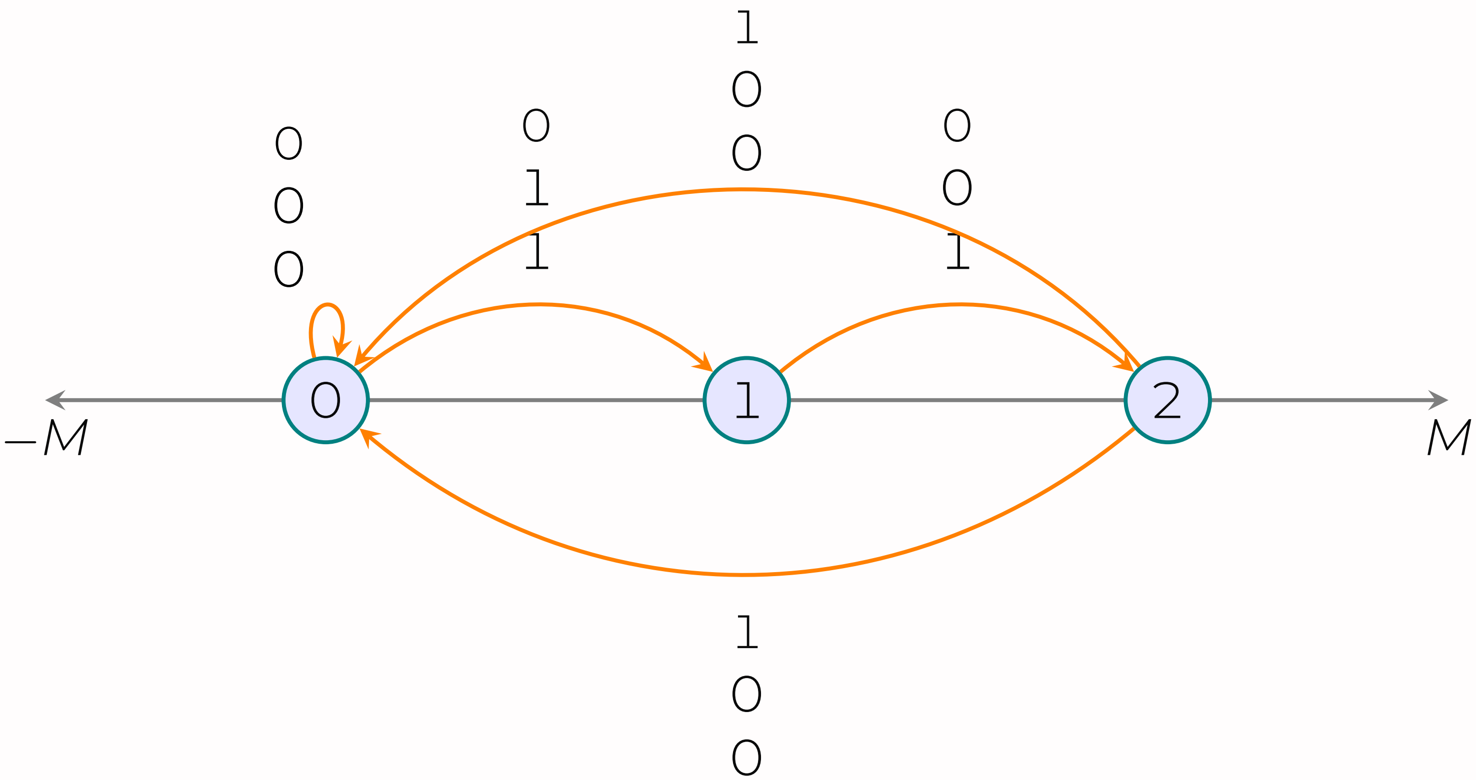
$$x \to 10001(17)$$

$$y \to 00100(04)$$

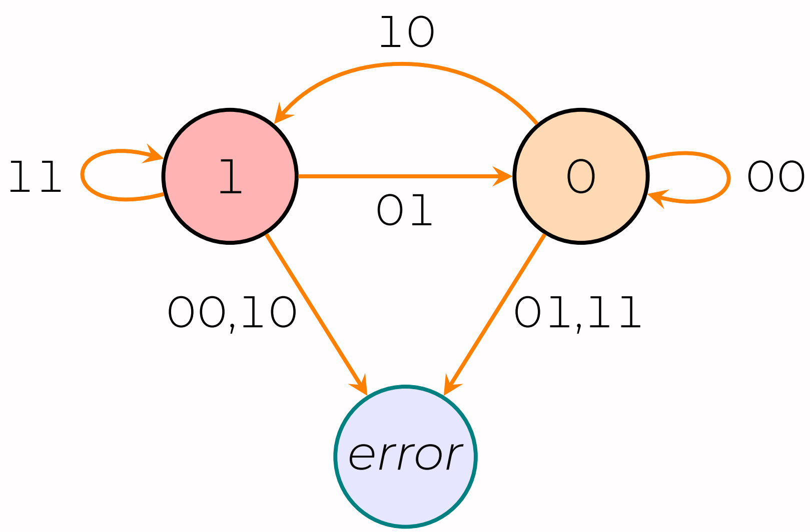
$$z \to 01100(12)$$

Keep track of the weighted sum required in the future to achieve the original weighted sum of b.

Automata Transitions



Construction of automata for $y = 2x + 1$



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Abstract

Key Insights

- An important class of subsets of lattice points in $\mathbb{N}^n$ arises from the theory of context-free languages — the semilinear sets.
- A set is semilinear if it is a finite union of cosets of finitely generated sub-semigroups of $\mathbb{N}^n$.
- These sets are shown to be equivalent to those definable by modified Presburger formulas.
- A characterization is provided for which semilinear sets correspond to languages (via Parikh images).
- Using this, a decision procedure is derived to determine whether a given linear set corresponds to a context-free language.

Algebraic Structures in $\mathbb{N}^n$

- Lattice Points: Elements of $\mathbb{Z}^n$. We focus on the subset $\mathbb{N}^n$ (nonnegative coordinates).
- Monoid: A semigroup with an identity element. $\mathbb{N}^n$ under vector addition is a commutative monoid with identity 0.
- Semigroup: A set S with an associative binary operation. Here: $S \subseteq \mathbb{N}^n$, closed under addition.
- Finitely Generated Semigroup: Exists $v_1, \dots, v_k \in \mathbb{N}^n$ such that:

$$S = \left\{ \sum_{i=1}^{k} a_i v_i \mid a_i \in \mathbb{N} \right\}$$

  Closed under addition by construction.
- Subsemigroup: $T \subseteq S$ that is also a semigroup (closed, associative).

Semilinear Sets in $\mathbb{N}^n$

- Coset (of Semigroup): For $a \in \mathbb{N}^n$, and S a semigroup:

$$a + S = \{a + s \mid s \in S\}$$

- Semilinear Set: A finite union of cosets of finitely generated semigroups:

$$\bigcup_{i=1}^{m} (a_i + \langle v_{i1}, \ldots, v_{ik_i} \rangle)$$

- Properties:
  - Closed under union, intersection, and complement (Boolean operations).
  - Every semilinear set is definable in Presburger arithmetic.
  - Intersection of two finitely generated semigroups is again finitely generated.

Semilinear Sets: Internal Characterization

Internal Structure

- A semilinear set is a finite union of linear sets.
- Each linear set is defined by a constant $c \in \mathbb{N}^n$ and a finite set of periods $P \subseteq \mathbb{N}^n$.
- The set consists of all points of the form:

$$c + \lambda_1 p_1 + \cdots + \lambda_k p_k \quad (\lambda_i \in \mathbb{N})$$

- This is an internal description constructed using additive generators and base points.

Equivalence with Presburger Descriptions

Equivalence Theorem (Section 1)

- The class of semilinear sets is identical to the class of sets defined by modified Presburger formulas.
- Each internal (semilinear) description can be effectively translated into an external (Presburger) formula, and vice versa.
- More precisely, each set in the class is defined as the extension of a modified Presburger formula with n free variables, i.e., the set of all n-tuples of nonnegative integers satisfying the given formula.
- This mirrors a classical analogy in linear algebra:
  - Internal: A finite set of vectors that spans a subspace
  - External: A finite system of linear equations whose solution set defines the subspace

Connection to Formal Languages

Semilinear Sets and Language Theory

- Semilinear sets arise naturally as Parikh images of formal languages.
- Our interest in semilinear sets stems from their relation to languages. Part III is devoted to this relation. It is shown that those semilinear sets which correspond to languages can be given semilinear descriptions of a particular form.
- For linear sets, a decision procedure is provided for determining whether an arbitrary set corresponds to a language.

Linear Sets in $\mathbb{N}^n$

Semigroup Structure

Let $\mathbb{N}$ be the set of nonnegative integers, and let $\mathbb{N}^n$ be the Cartesian product of $\mathbb{N}$ with itself n times.
For $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n)$ in $\mathbb{N}^n$:

$$x + y = (x_1 + y_1, \dots, x_n + y_n)$$

For $t \in \mathbb{N}$:

$$t \cdot x = (tx_1, \dots, tx_n)$$

Then $\mathbb{N}^n$ forms a commutative semigroup (under addition), and a partial order is defined by:

$$x \leq y \iff x_i \leq y_i \text{ for all } 1 \leq i \leq n$$

Linear and Semilinear Sets

Linear Sets

Given subsets $C, P \subseteq \mathbb{N}^n$, define:

$$L(C; P) = \{x_0 + x_1 + \cdots + x_m \mid x_0 \in C,\ x_i \in P\}$$

This describes all elements generated by adding a constant x_0 and a finite sequence of periods from P.
If $C = \{c\}$ and $P = \{p_1, \dots, p_r\}$, we write:

$$L(c; P) = L(\{c\}; P)$$

A subset $L \subseteq \mathbb{N}^n$ is called linear if $L = L(c; P)$ for some $c \in \mathbb{N}^n$, $P \subseteq \mathbb{N}^n$.

Semilinear Sets

Linear and Semilinear Sets

Let $P \subseteq \mathbb{N}^n$. The set P generates a finitely generated sub-semigroup $S \subseteq \mathbb{N}^n$, and for any $c \in \mathbb{N}^n$, the set:

$$L(c; P) = \left\{ c + \sum_{i=1}^{k} a_i p_i \mid a_i \in \mathbb{N},\ p_i \in P \right\}$$

is a coset of S in $\mathbb{N}^n$ containing c.
Thus, a set $L \subseteq \mathbb{N}^n$ is linear if and only if it is a coset of a finitely generated sub-semigroup of $\mathbb{N}^n$.

Semilinear Sets

Semilinear Sets

A subset of $\mathbb{N}^n$ is called semilinear if it is a finite union of linear sets.

A Non-Semilinear Sub-semigroup in $\mathbb{N}^2$

Definition

Let:

$$X = \{(x, y) \in \mathbb{N}^2 \mid y \leq x^2\}$$

A Non-Semilinear Sub-semigroup in $\mathbb{N}^2$

Definition

Let:

$$X = \{(x,y) \in \mathbb{N}^2 \mid y \leq x^2\}$$

Every vertical line $x = k$ intersects X in a finite set:

$$X \cap \{k\} \times \mathbb{N} = \{(k,y) \mid y \leq k^2\}$$

Thus, X cannot contain any linear set with a period of the form $(0,y)$.

Why X is Not Semilinear (1)

Assumption

Suppose:

$$X = \bigcup_{i=1}^{m} L(c_i; P_i)$$

where each $L(c_i; P_i)$ is a linear set.
Since each vertical line intersects X in a finite set, no period $p \in P_i$ can be of the form $(0, y)$.
Hence, all periods must satisfy $x > 0$. Let:

$$M = \max\left\{ \frac{y}{x} \;\middle|\; (x, y) \in \bigcup P_i,\ x > 0 \right\}$$

Why X is Not Semilinear (2)

Contradiction via Slope

Pick two points in X: (z_1, z_1^2), (z_2, z_2^2) with $z_1, z_2 > M$, where

$$M = \max \left\{ \frac{y}{x} \;\middle|\; (x,y) \in \bigcup P_i,\ x > 0 \right\}$$

Then the slope between them is:

$$\frac{z_2^2 - z_1^2}{z_2 - z_1} = z_2 + z_1 > 2M$$

which exceeds all period slopes.

Thank You!

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